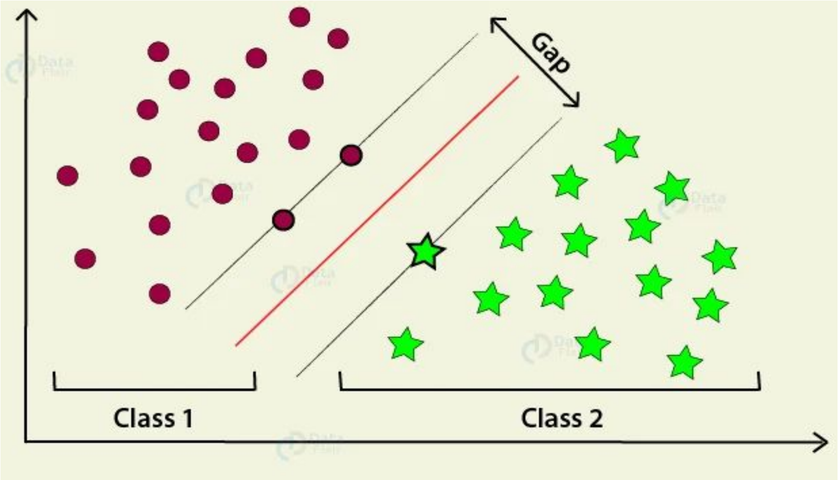
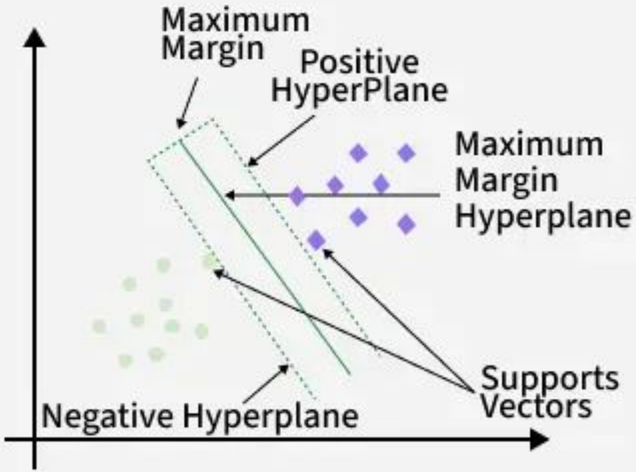
Support Vector Machine (SVM)

- Is a supervised machine learning algorithm used for classification and regression tasks. It tries to find the best boundary known as hyperplane that separates different classes in the data.
- It is useful when you want to do binary classification like spam vs. not spam or cat vs. dog.
- The main goal of SVM is to maximize the margin between the two classes. The larger the margin the better the model performs on new and unseen data.



Support Vectors & Hyperplane

- Support Vectors are the closest data points to the hyperplane that define the class boundary.
- A hyperplane is a plane that separates different classes.



Key Concepts of Support Vector Machine

- **Hyperplane**: A decision boundary separating different classes in feature space and is represented by the equation wx + b = 0 in linear classification.
- **Support Vectors**: The closest data points to the hyperplane, crucial for determining the hyperplane and margin in SVM.
- **Margin**: The distance between the hyperplane and the support vectors. SVM aims to maximize this margin for better classification performance.
- **Kernel**: A function that maps data to a higher-dimensional space enabling SVM to handle non-linearly separable data.
  - **Linear Kernel**: For linear separability.
  - **Polynomial Kernel**: Maps data into a polynomial space.
  - **Radial Basis Function (RBF) Kernel**: Transforms data into a space based on distances between data points.
- **Hard Margin**: A maximum-margin hyperplane that perfectly separates the data without misclassifications.
- **Soft Margin**: Allows some misclassifications by introducing slack variables, balancing margin maximization and misclassification penalties when data is not perfectly separable.

SVM Numerical Example (Real-life Dataset)

Problem

A bank wants to classify whether a loan should be approved based on:

- Monthly income (x_1)
- Credit score (x_2)

Class labels:

- −1 → Reject loan
- +1 → Approve loan

Dataset

Income (x_1)	Credit Score (x_2)	Loan
2	3	−1
3	2	−1
6	6	+1
7	5	+1

(Values are scaled for simplicity.)

Step 1: Hyperplane equation

SVM decision boundary:

$$w_1x_1 + w_2x_2 + b = 0$$

Step 2: Support vectors

The closest points between the two classes are:

Reject class → (3,2)

Approve class → (6,6)

For support vectors:

$$w \cdot x + b = -1$$

$$w \cdot x + b = +1$$

Step 3: Form equations

For (3,2), y = −1:

$$3w_1 + 2w_2 + b = -1$$

For (6,6), y = +1:

$$6w_1 + 6w_2 + b = 1$$

Step 4: Solve

Subtract equations:

$$(6w_1 + 6w_2 + b) - (3w_1 + 2w_2 + b) = 2$$

$$3w_1 + 4w_2 = 2$$

Assume equal contribution from features:

$$w_1 = w_2 = 0.3$$

Substitute back:

$$3(0.3) + 2(0.3) + b = -1$$

$$0.9 + 0.6 + b = -1$$

$$b = -2.5$$

Final Decision Function

$$0.3x_1 + 0.3x_2 - 2.5 = 0$$

Step 5: Prediction example

New applicant:

Income = 5

Credit score = 5

$$0.3(5) + 0.3(5) - 2.5$$

$$1.5 + 1.5 - 2.5 = 0.5$$

Since result > 0:

Loan Approved (+1)

Interpretation

SVM learned:

- Higher income + credit score → approval
- Lower values → rejection
- Boundary maximizes separation margin